TEACHER GUIDE · GRADE 2 · EVENTS & PARALLEL SCRIPTS

When I Receive

Quick facilitation notes & answer key

Goal

Students trace the ORDER that event-triggered scripts run in. Bit (green flag) broadcasts 'start'; Pixel ('when I receive start') waits, so it runs second. This links the receive event to sequencing — the message decides who goes first and who goes next — which is exactly the 'sequential and concurrent events' language of Grade-2 C3.1/ C3.2.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: two students; Bit moves and says 'start!', then Pixel may move. Ask the class 'who went first? who went second? why?'

Run it (about 20 min)

1. I DO: read Bit — 'green flag, 1 forward 3 → 4, then broadcast start.' Bit runs first.
2. WE DO: read Pixel — 'when I receive start'. 'Pixel waited; it runs second when start arrives: 0 forward 2 → 2.'
3. YOU DO: students name first (Bit/green flag), second (Pixel/receive start), and both landings.
4. Connect: 'The message sets the order — sender first, receiver next.'

Answer key (modelled by the run-gate)

Ex 1 — Bit runs first; its event is 'when green flag clicked'.

Ex 2 — Pixel runs second; it starts 'when I receive start' (the message Bit broadcast).

Ex 3 — Bit: 1 → forward 3 → 4. Pixel: 0 → forward 2 → 2. Bit lands on 4, Pixel on 2.

Order note: even though the green flag is one event, Pixel's start is delayed until the broadcast arrives, so the run order is Bit then Pixel.

Watch for & differentiate

Common error: saying both start on the green flag — only Bit has the green-flag hat; Pixel waits for the message.

Common error: mixing up the homes — Bit starts at 1, Pixel at 0; read each hat's sprite carefully.

Simplify: number the two scripts 1 and 2 in run order before finding landings.

Extend: ask what the order would be if Pixel also broadcast a message at its end to a third script (sender → receiver → next receiver).

Success indicator: the child names Bit first / Pixel second and lands Bit on 4 and Pixel on 2.